AYUSH KATIYA

Robotics | Autonomous Systems | Computer Vision | Controls | Intelligent Manufacturing

Tempe, AZ | akatiya@asu.edu | (623) 845-2525 | linkedin.com/in/ayushkatiya
github.com/AyushKatiya20322 | Portfolio

Education

Arizona State University Tempe, AZ
M.S. Robotics and Autonomous Systems - AI Specialization *Expected MAY 2027*

Relevant Coursework: Robotic Systems I & II, Perception in Robotics, Artificial Intelligence, Advances in Robotics, Applied Linear Algebra

Technical Skills

Languages: Python, C/C++, MATLAB, SQL
Robotics & Autonomy: ROS 2, OpenCV, YOLO, Computer Vision, Object Detection, Path Planning, Robotic Manipulation, Kinematics
AI/ML: LLM/VLM Systems, Reinforcement Learning, CNNs, Vision-Language Grounding, Depth Estimation
Controls & Hardware: Simulink, PID/PI Control, Arduino, Sensors, Actuators, Data Acquisition, System Integration
Industrial & Engineering: PLC/HMI, Ignition, OPC UA, Modbus, MQTT, REST APIs, SolidWorks, CAD
Developer Tools: Linux, Git/GitHub, Postman, Jupyter

Research Experience

Center for Bio-mediated & Bio-inspired Geotechnics (CBBG) Arizona State University
Research Aide *2026 – Present*

- Conduct experimental research involving **sensing, instrumentation, data acquisition, and engineering analysis** for bio-inspired and geotechnical systems.
- Integrate laboratory hardware and sensors, process experimental data, document procedures, and support repeatable experimental workflows.

School of Manufacturing Systems & Networks Arizona State University
Volunteer Researcher *2026 – Present*

- Develop and test robotics and intelligent-manufacturing systems involving **automation, sensing, robotic hardware, and experimental validation**.
- Integrate mechanical, electrical, and software subsystems and perform system-level testing for robotics research platforms.

Selected Robotics & AI Projects

Language-Driven Autonomous Robotic Manipulation *ROS 2 | Python | LLM/VLM | Computer Vision*

- Built an end-to-end robotic manipulation pipeline translating **natural-language commands into perception, task planning, motion planning, grasping, and physical execution**.
- Integrated vision-language grounding, monocular depth estimation, spatial reasoning, and agentic task decomposition for autonomous object selection and manipulation.

Autonomous Vision-Based Maze Solver *Python | OpenCV | Gemini | Dobot*

- Developed a perception-to-control pipeline to extract maze geometry, determine collision-free routes using AI-assisted reasoning with **A*/BFS fallbacks**, and execute trajectories on a Dobot robot.
- Implemented homography-based image-to-robot coordinate calibration, path validation, trajectory smoothing, and execution safety checks.

Real-Time Object Detection & Robotic Palletizing *Python | YOLO | Dobot Magician Lite*

- Integrated **real-time YOLO object detection** with robotic manipulation to identify, classify, pick, and sort objects using a suction-gripper end effector.
- Developed perception-to-action logic and robot motion sequences for autonomous, repeatable pick-and-place operation.

Research Publication

Katiya, A. et al. "Enhancing Food Type Recognition: A Comprehensive Study on Sequential Convolutional Neural Networks for Image Classification Accuracy," *Qeios*, 2024. Developed a CNN-based Food-101 image-classification system achieving **89.84% accuracy**. doi:10.32388/UFFBSR.2

Certifications: MATLAB, MathWorks (2025) | Simulink, MathWorks (2025)